

M002 Extension 2 — Piston & Logic Maze

Topic: Two Pistons + AND Conditions · **Course:** Maze Madness · **Level:** Extension (Advanced) · **Time:** about 60 minutes

This is an **extension challenge** for students who finished the weekly mazes. The agent follows a **redstone trail** through a long maze. At some junctions it must check **two things at once** — an **AND** condition — to pick the right way. Twice along the path, a **lever powers a piston** that pulls a block back and opens the gate. Your job: drive the agent through the **whole maze**, start to finish.

Chat words: **l** turn left, **r** turn right, **run** start the solver, **rl** teleport the agent back to you.

1 · Read the Chat Commands

Each block binds a **chat word** to an action.

```
def on_chat():
    agent.turn(RIGHT_TURN)
    player.on_chat("r", on_chat)
```

What does the agent do when you type **r**?


```
def on_chat():
    agent.teleport_to_player()
    player.on_chat("rl", on_chat)
```

You type **rl**. Where does the agent go, and when does that help in a long maze?

This maze has two pistons far apart. Why test the first half with **l** and **r** before you type **run** ?

2 · Trace the Solver 🔍

This loop runs when you type `run`. It follows the trail, makes AND decisions, and flips a lever at each piston gate.

```
while True:
    agent.move(FORWARD, 1)
    if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)
        agent.move(FORWARD, 1)
    elif agent.detect(BLOCK, AHEAD) and agent.detect(REDSTONE, RIGHT):
        agent.turn(RIGHT_TURN)
        agent.move(FORWARD, 1)
    elif agent.detect(BLOCK, AHEAD):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

The loop is `while True`. When does it stop on its own?

First check: `detect(REDSTONE, LEFT)` and `detect(REDSTONE, AHEAD)`. Describe a spot in the maze where both are true.

The last `elif` is a piston gate: a block ahead, no redstone beside it. What does `agent.interact(AHEAD)` do, and why can the agent move forward right after?

Why does check order matter? What breaks if `elif agent.detect(BLOCK, AHEAD)` ran first, before the two AND checks?

3 · Spot the Bug 🐛

Each block is broken. Read its goal, rewrite it, then explain the fix.

Bug A — Turn left **only when redstone is left AND ahead**. This turns on either one.

```
if agent.detect(REDSTONE, LEFT) or agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)  
    agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong?

Bug B — The solver should check on **every step** for the whole maze. This checks once and stops.

```
agent.move(FORWARD, 1)  
if agent.detect(BLOCK, AHEAD):  
    agent.interact(AHEAD)  
    agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong?

Bug C — At a piston gate, **flip the lever first**, then cross the gap. This walks forward before powering the piston, so it hits a closed wall.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(AHEAD)
```

Fixed code:

Why was it wrong?

4 · Make It Your Own

Open the **M002 EXT 2** world and run `run`. Now **change one thing** and predict before you test.

Pick **one** (or your own):

- Add a chat word `back` that turns the agent around (two right turns).
- Place a block (`agent.place(...)`) each time it flips a lever, so you can see how far it got.
- Add an `elif` for a junction where redstone is on the **right and ahead** at once.

Which did you pick? Write the code.

Predict: what will happen?

Test it. What actually happened?

run

Send a photo or video of the agent at the goal, then explain. Use these starters — 4 to 6 sentences.

The maze had two pistons, so the agent had to ...

At a junction I used an AND condition to ...

At a piston gate, the agent flipped the lever first because ...

The change I made in section 4 was ...

The hardest part was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Film on a phone while the agent solves the whole maze. Teach it like the viewer has never seen it. Try to use: **detect, redstone, trail, AND, piston, lever, loop.**

1. Show the maze and redstone trail before you start.
2. Type **run** and show the agent following the trail.
3. Read one AND check out loud and say what it decides.
4. Show the agent reaching a piston gate and flipping the lever.
5. Show it reaching the goal.

Write what you will say. Plan it here first — read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.